

Connected we stand: Lead firm ownership ties in the global petrochemical industry

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ABSTRACT

Using oil, gas, and coal to produce platform chemicals on an enormous scale, the petrochemical industry constitutes a core part of the global energy order. Given demand growth for petrochemicals, the sector is set to become increasingly important to fossil fuel interests. Arguing that internationalised networks help structure the social metabolism and are important for transformative change, this paper sets out to analyse economic ties in the global petrochemical industry. In this paper, we conceptualise such relations and explore how they foster alignment on a global scale. We emphasise the role of internationalised networks in global socio-technical regimes, arguing for the importance of economic ties that establish a financial and juridical relation. On this basis, we theorise that extensive lead firm ties strengthen global regimes and shape socio-technical reconfigurations to align the interests of incumbent actors. Applying this framework, we analyse ownership relations amongst lead firms in the global petrochemical sector. We find a polycentric but global network aligning interests across major producers which we argue helps maintain and reproduce commitments to fossil fuels. The findings illustrate the relevance of pursuing parallel transitions along the petrochemical value chain, including energy, chemicals, and plastics, to break from fossil fuel dependency.

1. Introduction

A substantial share of the social metabolism revolves around the production of synthetic materials, processes of which the petrochemical industry is at the heart. To produce chemicals, the petrochemical industry relies on fossil resources, including coal, oil, and gas, not only as sources of energy but also as feedstock. Despite being the industrial sector with the highest energy demand, petrochemicals have often been overlooked in both research and policy related to the energy and climate nexus (cf. Hanieh, 2021; IEA, 2018; Svensson et al., 2020; Wesseling et al., 2017). This omission is troubling, as the recent decade has seen massive investments into petrochemicals contributing further to escalate an already strong commitment to fossil-based chemical production (Bauer and Fontenit, 2021; VCI, 2021). In addition, several analyses project petrochemicals to be the main driver of oil demand growth in this decade (BP, 2022; IEA, 2018, 2021a). This outlook makes the petrochemical sector key to energy transitions and the struggle over the future of the global social metabolism. The potential for escalating commitments to fossil fuels (Arbuthnott and Dolter, 2013) through

petrochemicals highlights the need to assess the relations (financial, economic, and judicial) that underpin the maintenance of and investments in fossil capital.

To understand transition dynamics in industries on a worldwide scale, scholars in the field of sustainability transitions have invoked the concept of global regimes. Conceptualised as dominant institutional rationalities, global regimes are said to have reached validity beyond specific geographies (Fuenfschilling and Binz, 2018). The global regime concept helps theorise change dynamics on a global scale, highlighting how and why industries maintain socio-technical stability. For regime (re-)enforcement and (re-)enactment, Fuenfschilling and Binz (2018) highlight the importance of internationalised networks and their role in diffusing rationalities. In this paper, we seek to unpack such relations to better understand what makes regimes 'stay' (Turnheim and Sovacool, 2020). Focusing on the form and extensiveness of ties amongst actors orchestrating the global energy metabolism, we point to the risk of (re-)enforcement of carbon lock-in (Könnölä et al., 2006; Unruh, 2000) with important implications for the prospects of transformative change on a global scale.

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Starting from a relational perspective, this paper explores the potential role of internationalised producer networks in creating and upholding stable configurations, thereby maintaining the stability of the global regime in the petrochemical industry. Specifically, we conceptualise the importance of internationalised networks around the neo-Gramscian notion of *trasformismo*, arguing that economic ties between central actors in the global regime commit incumbents to pursue strategies of accommodation and absorption. Our empirical analysis explores ownership interlocks across the largest petrochemical producers, assessing economic integration on a global scale. In assessing integration across lead firms in the chemical sector, we argue that extensive economic ties in the sector work to support past and planned investments in fossil capital at a time when the industry as a whole is under pressure. We thereby add to recent research that has begun to unpack networks in the petrochemical industry (Buch-Hansen and Henriksen, 2019; Buch-Hansen and Larsen, 2021; Verbeek and Mah, 2020) and the social metabolism of petrochemicals more generally (such as Bauer and Fontenit, 2021; Hanieh, 2021; Mah, 2021a; Tilsted et al., 2022). While the analysis does not seek to determine direct causal links between the network and specific outcomes, we argue that it is relevant to understand the network structure as conditioning and possibly inhibiting radical industry transformation.

The paper continues as follows: Section 2 introduces the global regime concept and reviews the notion of global networks in the literature that the regime concept draws upon. Section 3 presents the methods and data used, as well as the empirical context, before turning to the empirical analysis. Section 4 discusses the findings and their implications, while section 5 concludes.

2. Global regimes and global networks

From an ecological economics perspective, understanding the economy as a metabolic organism, sustainability transitions extend beyond the techno-economical and involve changing social arrangements and relations of power that shape the provisioning of key societal functions (Røpke, 2019). This multi-faceted understanding of socio-economic systems that drive resource consumption and pollution demands engagement with their technical as well as their political, social, and economic aspects, spanning across scales and geographies (Røpke, 2016).

To understand the relations that connect and shape the social metabolism on a global scale, scholars in the sustainability transitions literature have invoked the notion of global socio-technical regimes. In transition studies, regimes are said to form the 'deep structure' of socio-technical systems that fulfil societal functions (Geels, 2010). Regimes constitute the most strongly institutionalised socio-technical configuration and help explain inertia and stability (Karlton and Sandén, 2012), as regimes exert more power over actors than so-called niches, i.e., weakly institutionalised configurations (Fuenfschilling, 2019; Mörner et al., 2021). Regimes solidify through the co-evolution of society and technology as formal and informal institutional structures that guide behaviour and shape social practices develop alongside technologies (Fuenfschilling and Binz, 2018). Building on, inter alia, the global value chain (GVC) and the global production network (GPN) literature, the global regime concept then points to institutional rationalities that exist beyond specific territorial contexts.

Bringing together various perspectives, Fuenfschilling and Binz (2018, p. 739) define a global socio-technical regime as the dominant institutional rationality in a socio-technical system, which depicts a structural pattern between actors, institutions and technologies that has reached validity beyond specific territorial contexts, and which is diffused through internationalised networks.

From this definition alone, global regimes appear rather stable. Yet, these institutional rationalities are, following Meyer (1999), only semi-coherent and, following GPN, multi-scalar. As with non-global regimes, they are contested by different actors and constantly renegotiated and

re-produced. Moreover, following the GVC literature, actors' abilities to influence the dominant institutional rationality depend on their positions in international networks as well as their capacities to invoke institutional power (Fuenfschilling and Binz, 2018). As cultural-cognitive rationalities, regimes exert mimetic pressures but can also influence behaviour through regulative and normative pressures, and by materializing in technology. Due to the multi-scalar nature of global regimes, spatial variation is bound to arise through interplays across scales and regionally anchored regimes, challenging simplistic notions of technology transfer (Monstadt and Schramm, 2017).

Nuancing the global regime concept, Fuenfschilling and Binz (2018) distinguish between strong and weak global regimes, allowing for varying degrees of institutionalisation and structuration across sectors. Strong and weak regimes differ in terms of the scale and scope of diffusion and their resistance to alternative configurations. As various rationalities co-exist in most socio-technical systems, different institutional logics compete across scales. The regime thus denotes 'the semi-coherent assemblage of elements of various ideal-type rationalities which are most deeply institutionalized (...), while elements of other institutional rationalities are much more fluid, unstable, and thus less influential' (Fuenfschilling and Binz, 2018, p. 739). As institutional rationalities diffuse through and take hold in different regions with otherwise different preconditions (cultural, institutional, material), the strength of the regime increases.

2.1. Conceptual foundations of internationalised networks

Internationalised networks constitute important relational structures in global socio-technical systems. Through connecting actors across places, scales, and sectors, networks make up central, if not foundational, elements of the global political economy (Coe et al., 2008; Dicken et al., 2001). Yet, the conceptualisation of networks in transition studies often remains flat, i.e., it does not distinguish between and specify the importance of different kinds of relations. Regimes were in early work seen as involving multi-actor networks (e.g. finance, research, and producer networks), with regimes being produced and reproduced by networks of various actors and institutions (Geels, 2002; Smith et al., 2005). Therefore, networks are said to be a generic feature of innovation systems (Grillitsch et al., 2019; Musiolik et al., 2012). Later work has described competing socio-technical configurations including both social and technical elements as networked with dominant paradigms constituting stable networks and niche innovations represented as emerging networks (Heiberg et al., 2022; Mörner and Binz, 2021). Neither of these abstractions, however, is particularly helpful in identifying and operationalizing internationalised networks that underpin the global regime, nor do they give emphasis to specific types of networks over others. To further unpack global networks, we therefore now turn to two of the approaches that the global regime concept draws from, namely the GVC and the GPN literatures.

Drawing from social network theory (Borgatti et al., 2018; Cronin, 2016; Robins, 2015, pp. 17–38), we focus on three aspects relevant to transitions. These include i) network conceptualisation, ii) collaboration, and iii) stability. *Conceptualisation* relates to the basic understanding of networks, in what sense the term is evoked, and the context and relations it is meant to describe. *Collaboration* pertains to the form and dynamics of network relations, e.g., the possibility of both collaborative and competitive ties as well as the co-evolution of trust, control, and learning in actor relations (Inkpen and Currall, 2004). Finally, *stability* refers to the role of networks in enabling stability and endurance in the network environment. Although we here delineate between different elements of network conceptualisation, these categories overlap. For example, collaboration in a network relates to how ties are defined. Thus, we do not see the categories above as fundamentally distinct but rather as analytical aids useful for review.

2.1.1. Networks in GVC and GPN

GVCs are, in essence, conceptualised as chain structures tied through dyadic inter-firm links, representing economic activity in a rather parsimonious manner (Bair, 2008; Dallas et al., 2019; Gibbon et al., 2008; Henderson et al., 2002). These relationships and their degree of formalisation within value chains exist on a spectrum, ranging from ‘hierarchies’ between headquarters and subsidiaries in an MNC to ‘markets’ with arm-length relations (with ‘captive’, ‘relational’ and ‘modular’ constituting the categories in between) (Gereffi et al., 2005). In line with the attention given to the production, distribution, and consumption of specific products and services in particular industries, the multinational corporations (MNCs) around which production networks are coordinated, i.e., lead firms, are of special importance to GVCs (Dallas et al., 2019; Yeung and Coe, 2015). Following a broadening in the GVC literature towards multipolarity, the importance of actors outside the core inter-firm networks has been flagged, and so has lead firm heterogeneity (Dallas, 2015; Nadvi and Raj-Reichert, 2015). This is closer to the core conceptualisation of production networks in the GPN literature, which are conceptualised as fragmented, complex, and operating across locations at a continuum of scales¹ (Coe et al., 2008; Dicken et al., 2001; Henderson et al., 2002). There is thus attention to extra-firm dynamics, institutions, and regionality in the core conceptualisation² (Coe and Yeung, 2015). Recent efforts to further develop the definition of GPNs see them as lead-firm-based configurations (Coe and Yeung, 2019) or as ‘chains’ of ‘multiple, differentiated networks at each stage of a given production network’ (Stephenson and Agnew, 2016, p. 559).

Regarding collaboration in GVCs, the aforementioned governance structure as well as the degree and form of power asymmetries shape the nature of relations. Thus, although participation in a given GVC is a way to acquire both knowledge, access, and assets (Taglioni and Winkler, 2016), lead firms typically decide the terms upon which inclusion is negotiated (Dallas et al., 2019; Gibbon and Ponte, 2005; Humphrey and Schmitz, 2002). Although the focus in GVC analysis initially tended to focus on coercive dynamics in relations, the approach now includes unintentional and more subtle dynamics which structure relations such as mimetic pressures through norms, conventions, and worldviews, similar to the regime concept (Dallas et al., 2019; Mondliwa et al., 2021). To add to the notion of inter-firm relations as power struggles, the GPN tradition emphasises dependency and aspects of mutual interest in networks (e.g., in the case of knowledge transfer). In this perspective, specific production networks configured around lead firms are understood as interfirm partnerships that collaborate and coevolve in competition with other networks (Yeung and Coe, 2015).

In terms of stability, a key tenet of GVCs is that the power to influence value chains mainly resides within lead firms (Dallas et al., 2019). Through their structural positions, lead firms have influence over and the ability to shape operations elsewhere in the chain (Gereffi et al., 2005; Ponte and Sturgeon, 2014). Questions of stability in inter-firm relations are, in this sense, matters of GVC governance and the strategizing of various actors, especially lead firms (Gibbon et al., 2008). Power dynamics thus help determine stability in GVCs. The extent to which certain forms of power are more resistant to change remains an open question and is highlighted as a point for future research by Dallas, Ponte and Sturgeon (2019, p. 678), who nevertheless point to the tendency for networks to generate monopoly power through dynamics of path dependency and lock-in. Complementing this perspective, the GPN approach highlights political-economic ‘drivers’ that actors face to explain change (and thereby stability). These important drivers are

different but related to the capitalist dynamics of cost-capability ratios, pressure to develop new markets, and financial discipline (Coe and Yeung, 2015; Yeung and Coe, 2015).

2.1.2. Lead firms in global regimes

Table 1 summarises the GVC and GPN network conceptualisations as presented above, which we use as a starting point to operationalise internationalised networks. While the GPN concept underscores the complexity of the internationalised networks that underpin global regimes, these perspectives help bring clarity by guiding our attention towards particular actors (lead firms) and particular relations (formalised economic ties). As leading incumbents in an industry, we posit that lead firms must, by definition, relate to the global regime, i.e., the globally dominant institutional rationality. At the same time, lead firms are in a position to reach out and relate to both local and global niches and regimes in different industries through various governance arrangements. For example, incumbent petrochemical companies have engaged in the production of bioplastics through joint ventures (Total, 2016; Tullo, 2011) and acquired smaller firms engaged in niche development of new products and processes (e.g. Evonik, 2021). Despite elements of mutual interests and dependency, the more asymmetric and hierarchical the power relation between lead firms and their respective counterparts, the more likely lead firms are to dictate (the form of) change within a given value chain. For example, MNCs have used the sustainability agenda for ‘green capital accumulation’, operationalizing sustainability in specific ways and squeezing suppliers to ensure new opportunities for accumulating capital (Ponte, 2020). In short, lead firms ‘navigate the storm’ from a structurally dominant position (van Mossel et al., 2018).

With this focus on lead firms and formalised inter-firm relations in mind, we now home in on the question of how such actors and ties in internationalised networks influence stability in global regimes in the face of transition pressures. To address this issue and put forward a coherent framework, we resort to neo-Gramscian theory and the notion of ‘trasformismo’.

2.2. Shaping socio-technical reconfigurations

In recent years, various scholars have pointed to the merits of neo-Gramscian perspectives on transitions (Ford and Newell, 2021; Szabo, 2022; Tilsted et al., 2022). In particular, Ford and Newell (2021) unfold the range of relevant neo-Gramscian concepts and their application to transition studies (and the widely invoked multi-level perspective), highlighting the relevance of trasformismo. Distinguishing between transitions and trasformismo, Newell (2019) defines the latter as

Table 1

Overview of the network conceptualization in the GVC and the GPN literature.

Scholarly tradition	Network definition and conceptualisation	Collaboration	Stability
Global value chains (GVC)	Inter-firm networks focused on dyadic ties between lead firms and suppliers at the industry level.	Nature of collaboration is shaped by power struggles. Lead firms typically decide terms on which suppliers access knowledge and assets through GVCs.	Stability hinges typically on lead firms and the form of GVC governance.
Global production networks (GPN)	More elastic network concept, emphasising complexity, fragmentation, and multiple scales.	In addition to power, GPN also emphasises dependency and mutual interests.	No a priori stance on what power dynamics lead to stability. GPN points to capitalist dynamics to explain the evolution of GPNs.

¹ While distinctions between the GVC and the GPN literatures have become less discernible in later years (cf. Coe and Yeung, 2019; Grabs and Ponte, 2019), the core network conceptualizations differ.

² The broad conceptualization of GPNs has also been a point of criticism as ‘the elasticity of this view of networks means that it often includes just about everything and lacks analytical boundaries and clarity’ (Sunley, 2008, p. 8).

the ability to accommodate pressures for more radical and disruptive change and to employ combinations of material, institutional and discursive power to ensure that shifts which do occur in socio-technical configurations do not disrupt prevailing social relations and distributions of political power (p. 28).

What *trasformismo* captures is that stability can take different forms, including that of accommodation. When transition pressures increase, be it at the regime or the niche level (or in combination), the ability of regime actors to absorb these pressures through coordinating and employing multiple forms of power (material, institutional, discursive) critically influences the prospects of change and the degree of stability in the system (Ford and Newell, 2021).

Material, institutional, and discursive power are related but point to different sources of influence and the multiple aspects of maintaining regime stability. Material power arises from controlling production and finance and from promoting and ensuring economic growth, a core objective of states (Ford and Newell, 2021). Institutional power, in turn, builds on this material power and is employed to influence and shape policymaking, both via formal and informal channels. Discursive power, lastly, is employed by actors to build consent around specific projects through story building and narrative work pushing, e.g., discourses of climate delay or certain policy narratives (Lamb et al., 2020; Palm et al., 2022). Regime actors in dominant positions, who hold power in all three dimensions, can seek to outright resist and inhibit change or, in the face of mounting pressure, pursue *trasformismo* to shape change processes (Ford and Newell, 2021; Newell, 2019). The prospect of change is then decided by not only whether socio-technical reconfiguration does occur but also what form such alterations take. This is not to say that genuine alternatives to the regime—or in neo-Gramscian terms, the historical bloc—are not possible. In transition studies as well as neo-Gramscian theory, regime actors are contested and their dominance remains contingent (Geels, 2014; Levy and Newell, 2002).

The process of accommodation encompasses strategies to foster consent and acquire the acquiescence of important or powerful actors and actor groups so as to shape socio-technical reconfigurations in desired directions (Ford and Newell, 2021; Levy and Newell, 2002). Accommodation happens both in response to niche and landscape pressures. As for landscape pressures, the overwhelming adaptation of growth-oriented and climate and circular economy discourse focusing on technical fixes in the wake of increased policy pressure illustrates such efforts (Mah, 2023; Markusson et al., 2017). In terms of niche pressures, incumbent actors can buy up, own, and occupy niche spaces, including alternative and potentially threatening technologies. By doing so, incumbents can demonstrate green credentials, build a knowledge base of competing configurations, and align niche industry advocacy organisations with the interests of regime actors (Ford and Newell, 2021; Ford, 2020; Haas, 2019). Ford and Newell (2021) also suggest that incumbents, through institutional power, can seek to influence policies that support niches or make compromises and concessions to avoid total reconfiguration and thereby sustain the regime.

By bringing the insights from this section together, we can address the issues of what forms of power and what types of relations that are relevant to foreground. Regarding the former, the neo-Gramscian perspective stresses material power as a foundation that other forms of power build on, which are employed in a coordinated fashion. In a time of mounting socio-ecological crises, these forms of power that enable not only resistance but accommodation, are critical for regime stability. Regarding the types of networks to home in on, we posit that internationalised networks amongst different lead firms warrant particular attention. Because ownership is hierarchical and represents a strong link between firms—as compared to other forms of inter-firm connections and value chain governance—lead firm ownership networks stand out. Such networks embody direct, dyadic forms of power but also reflect other and more diffuse forms of power that operate through mimetic pressures, as is characteristic of regimes. At the same

time, lead firm ownership networks do not represent specific value chains³ but rather concrete internationalised actor networks diffusing and reproducing global regimes. In that sense, we take inspiration from the more multi-faceted network concept found in GPN. The actors in lead firm networks take part in various GVCs (depending on the scope of their activities) and the inequalities that occur within, along, and through value chains (Lang et al., 2022).

Given the advantageous position of lead firms, making them uniquely capable of absorbing both global niche and broader landscape transition pressures, we argue that extensive lead firm networks help maintain global regime stability. Extensive corporate networks do not necessarily entail a non-changing system but might as easily foster *trasformismo*. Faced with different types of pressures, lead firm networks allow for alignment between incumbents in seeking to accommodate such pressures through different means without disrupting the underlying rationality. Extensive lead firm networks allow for wide-spread diffusion and can, following our argument above, increase the ability to absorb change pressures. This argument concerning lead firm networks and the ability of incumbent actors to accommodate and absorb pressures is illustrated in Fig. 1, which depicts an ideal-typical representation of *trasformismo* in a global regime with integrated lead firm networks facing external pressures.

In proposing that lead firm ownership networks are particularly worthy of note in relation to processes of change, we stress the importance of material, economic, and juridical relations. Ownership ties between lead firms are materially anchored and backed by judicial obligations. These ties are instances of collaboration and/or hierarchy and represent common commitments. An extensive lead firm ownership network ensures that firms have common material interests. Anchoring common interests in joint commitments to specific projects arguably fosters alignment and accommodation in response to pressures from outside the regime that challenge existing socio-technical configurations, i.e., *trasformismo*. This is not to say that lead firm ownership networks do not arise in already existing internationalised networks. Instead, in line with the sustainability transitions literature, we see lead firm networks as both an outcome and a facilitator of institutionalisation. Yet once in place, these relations underpin and condition the global regime that seeks to uphold a given socio-technical system.

3. Relations in the global petrochemical industry

3.1. An industry under pressure?

The petrochemical industry is characterized by a set of contradictions. Used across industrial processes for a variety of purposes, petrochemicals ensure the ubiquity of fossil fuels in modern society. While their broad use means that the chemicals are, as is often stressed by industry, needed in sustainability transitions for applications such as carbon fibre for wind turbines (Tilsted et al., 2022), the industry is ‘locked-in’ to fossil-based production (Janipour et al., 2020, 2022; Tilsted et al., 2023; Wu et al., 2015). Moreover, foreseeing that the ‘future of oil is in chemicals’ (Tullo, 2019), the industry is booming in terms of production capacity, fuelled especially by expansions in China, the Middle East, and the United States (Amghizar et al., 2017; Bauer et al., 2023b; Bauer and Fontenit, 2021). At the same time, actors across petrochemical value chains are facing increasing external pressures. The industry is criticised for its central role in a range of issues, from environmental justice (Davies, 2019; Davies and Mah, 2020; Ho, 2014), toxicity and health (Davies, 2018; Jephcote and Mah, 2019; Tickner et al., 2021), climate change, and environmental governance (King and Lenox, 2000; Mah, 2021b; Palm et al., 2022). Together, these

³ To understand specific GVCs, in-depth engagements and case studies are required (see for example Mondliwa et al. (2021) for an application of GVC theory to the case of the South African petrochemical value chain).

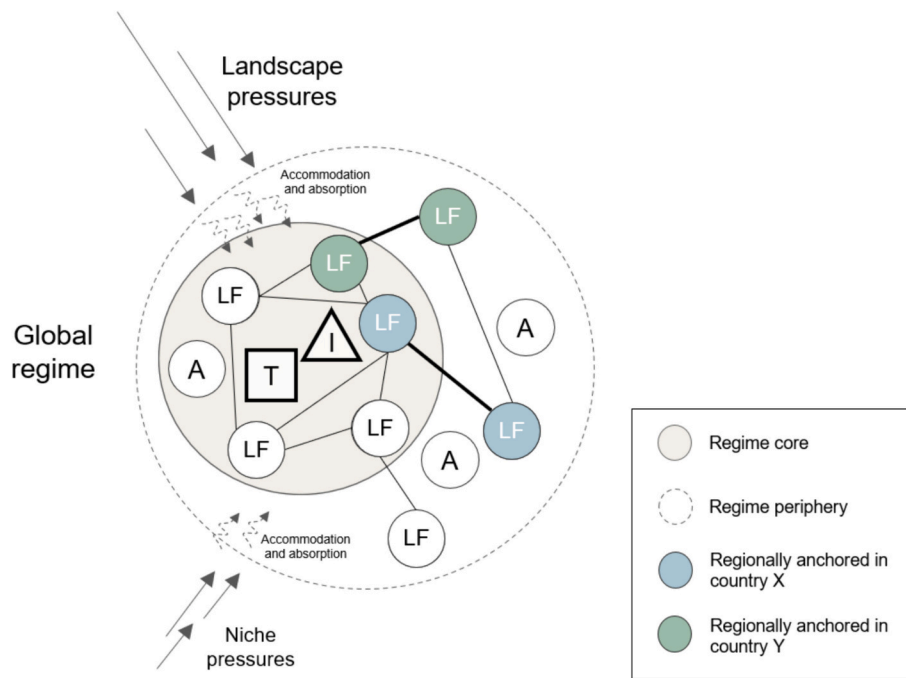


Fig. 1. Ideal typical representation of accommodation of transition pressures in an industry with strong lead firm networks. Own figure inspired by Fuenfschilling and Binz (2018), Miörner and Binz (2021), and Dallas et al. (2019). I – institutions, T – technologies, A – actors, LF – lead firms. Wider edges represent stronger, potentially more asymmetric LF relations.

contradictions portray an industry at a crossroads with mounting transition pressures.

The global chemical regime is, according to Bauer and Fuenfschilling (2019, pp. 175–176), centred around three core characteristics: 1) the historical linkages with energy production and the resulting integration of fuels and chemicals production in large petrochemical clusters as well as a common chemical engineering knowledge base (Rosenberg, 2000); 2) a research and development focus on process improvements due to a molecular similarity in products that rely on economies of scale (Ren, 2009) and are subject to downstream product requirements and cost-orientation; 3) the importance of traditional chemical companies, complemented by specialized engineering firms who develop and license process technologies (Arora and Fosfuri, 2000). Taken together, Bauer and Fuenfschilling (2019) emphasise stability as well as the dominance and permanence of incumbent firms when describing the structure of the industry. Further, they highlight the sector's global nature with global markets for both products and technologies, cumulative innovation patterns, and strong, long-standing MNCs. Over time, the pressure to sustain market development, coupled with a focus on decreasing costs through efficiency gains and scale, has been key to developments in the industry (Aftalion, 2001).

The petrochemical industry is arguably a good example of *trasformismo* and accommodation of landscape pressures. Over time, environmental discourses have influenced the chemical sector, which has gradually evolved to integrate aspects of 'green chemistry' and 'eco-innovation' (López and Montalvo, 2015). More recently, the marine plastics crisis has faced the industry, threatening its 'social license to operate' (Mah, 2021b, p. 129). Lead firms have responded by embracing the circular economy discourse, which, in a few years during the 2010s, was adopted by all the top firms, forming joint alliances and action groups (Mah, 2022, 2023). The way the circular economy discourse has been adopted by the industry, focusing on large-scale chemical recycling and marine plastic pollution prevention, however, is far from the calls for reducing the use of plastics and capping production made by other actors (Bergmann et al., 2022; Mah, 2021a, 2021b). Drawing on extensive field research and interviews with corporate stakeholders, Mah (2021b) documents how industry-level threats made the industry

come together, collaborating to contain pressures and use them to expand new markets.⁴ Moreover, climate-related transition narratives are prominent amongst petrochemical majors, positioning the industry as integral to sustainability efforts and a low-carbon future by realigning it with decarbonisation. Building on lobbying efforts and their structural importance, industry actors seek to influence pressures for sector-level reconfigurations (Tilsted et al., 2022).

In line with our argument laid out above, the globalized nature of the petrochemical industry and its engagement in *trasformismo*-like strategies means that we would expect internationalised lead firm networks. From a coevolutionary perspective, such networks both facilitate and result from alignment, pointing to self-enforcing feedback mechanisms. Strong lead firm networks would thus instill continued collaboration and accommodation going forward. The extent to which these relations are materially anchored in existing fossil-based investments, i.e., through common ownership, bears implications for the prospects of a global-level low-carbon reconfiguration of the petrochemical industry. Petrochemical lead firm networks represent a common commitment to fossil-based infrastructure. With investment horizons of several decades and high upfront investment costs (Wesseling et al., 2017), and with expected profits premised on maintaining and/or upscaling production, lead firms have massive committed emissions given current investment patterns (Bauer et al., 2023a; Bauer and Fontenit, 2021). The alternative, decommissioning or scaling down production, would mean that lead firms' commitments end up as stranded assets. To remain profitable, these corporations will continue to rely on fossil fuels. And where individuals can come and go, infrastructure stays. Thus, the industry is under financial pressure to continuously produce *petrochemicals* on a massive scale in the years to come.

At the same time, the 2020s is a crucial decade for investments in

⁴ In 2019, the corporately founded Alliance to End Plastic Waste was established with multiple petrochemical lead firms joining. Setting aside intensive rivalries, working to defend plastic markets and deter responsibility, the alternative to collaboration was 'leaving it to the regulators' (Mah, 2021a, p. 133). This alliance has been reported to be an outcome of coordinated work orchestrated by petrochemical actors (Baker et al., 2022).

fossil-based chemical production infrastructure. According to the International Energy Agency's Net-Zero Emissions by 2050 roadmap, carbon dioxide emissions from the chemicals industry must decrease by 2030 and be halved by 2040 from 2020 levels, while production volumes increase by around 30% (IEA, 2021b). Drastically reducing emissions on a global scale in an industry where 2050 is only one investment cycle away means that this decade constitutes a critical 'window of opportunity' (IEA, 2021c, p. 124). In the IEA net zero scenario, the share of total chemical production carried out via conventional routes is less than 90% by 2030, and all capacity additions feature 'near-zero' technologies (IEA, 2021c, pp. 121–131). But while the IEA scenario emphasises that refurbishments and lifetime extensions of existing industrial assets along conventional routes are not in line with global temperature goals, projections of planned and announced expansions show a 40% production capacity growth in this decade (Global Data, 2021; IEA, 2021c).

3.2. Methods and data

To analyse lead firm networks in the chemical industry, we restrict our analysis to the 52 largest companies globally⁵ (see Table 2). These firms vary significantly in global market share, but all play important roles regionally and most can arguably be characterized as lead firms. For example, in their analysis of the petrochemical value chain in South Africa, Mondliwa et al. (2021) characterise Sasol, which is amongst the companies with the lowest chemical sales in our analysis, as a lead firm. Following the above, we see lead firms as connected to the global regime. The list includes major chemical producers, covering MNCs from both the traditional petrochemical strongholds of Western Europe and the United States (e.g., BASF (DE) and Dupont (US)), firms that emerged in the latter half of the 20th century (Mitsubishi Chemical (JP) and Formosa Plastics (TW)), as well as the largest emerging market chemical companies (Sinopec (CH) and Reliance (IN)). Showcasing the relationship between chemicals and fossil fuel extraction, both international oil companies (e.g., ExxonMobil and Chevron Phillips) and national oil companies (e.g., Sinopec and Sasol) are part of the sampled MNCs.

We restrict our analysis to ownership networks, considering instances where business entities are listed as subsidiaries of more than one of the lead firms in the sample. We use the Orbis (Bureau van Dijk) database, which compiles data from a range of sources to present comprehensive information on companies worldwide. This means that the analysis presented below depends on the extensiveness and quality of global company databases (Heemskerk et al., 2018). Extracting the full lists of registered subsidiaries for all 52 firms produced a data set totalling 37,030 observations with 28,455 unique company IDs. This data set includes all subsidiaries of the sampled companies listed in Orbis as active at the time (July 2, 2021), including entities whose ownership shares are unknown and non-majority-owned subsidiaries. Thus, the data reflects the ties formalised in joint ownership of subsidiaries and captured in the database at that moment. The dataset allows for the creation of a network of parent MNCs and their subsidiaries, 1574 of which have interlocking ownership with a varying number of parent companies. Links between MNCs (indicating cross-ownership) were removed to produce a two-mode (or bipartite) network, which was then projected to a one-mode network configuration of only the parent MNCs, in which the edges are constituted and weighted by all instances of joint ownership of a subsidiary.

To visualise the network and calculate descriptive statistics, we use the software Gephi. For the network graph, we include parent companies as nodes, with edges representing instances of joint ownership.

We include all subsidiaries with ties to more than a single parent in the dataset. Tie width is scaled by the number of subsidiaries two MNCs have in common. Similarly, nodes are scaled by their eigenvector centrality, capturing the centrality of related nodes. Descriptive network statistics are available in Table 3 and node-specific statistics are available in Appendix A.

3.3. Lead firm ownership networks

3.3.1. An integrated network

The petrochemical lead firm ownership network illustrated in the network graph (Fig. 2) supports the characterisation of the petrochemical industry as globalized (cf. Section 3.1). 284 linkages exist across the network which is 21.4% of the total number of possible ties (see Table 3). Specifically, major petrochemical producers across all continents outside of Oceania with headquarters in 18 different countries are linked through ownership in subsidiaries located in 56 different countries. Outside of three corporations,⁶ all of the sampled companies are linked through ownership ties and have one or more subsidiaries in common (as of July 2021). Among the connected companies, no producer has more than three nodes between them and any other companies in the network. The degree count, i.e., the number of actors with which a given company is connected, varies across the network (see Fig. 3). On average, each petrochemical producer has ownership ties to around 11 other producers in the network through common ownership in 76 subsidiaries, illustrating linkages across the largest firms in the industry on a global scale.

Considering the most integrated firms, 27 of the sampled companies have ownership ties to 10 or more of their counterparts. Of these, Saudi Arabian SABIC has the highest degree count of 28. SABIC also has, by a substantial margin, the highest betweenness centrality in the network, meaning that it occupies the most important position in terms of relating other firms to each other through the shortest possible paths (Robins, 2015). For closeness and eigenvector centrality, capturing respectively the average distance to other nodes and the centrality of related actors, a range of firms ranks as the most central. This is evident in the network graph, where nodes are scaled by eigenvector centrality, showing a range of centrally placed firms, including large Japanese producers as well as the biggest North American players in the form of ExxonMobil, Dow, and LyondellBassell. Eigenvector centrality is relevant for regime diffusion because it captures to what extent nodes are connected to other central lead firms. Being linked to central actors in the network is likely to mean exposure to dominant rationalities and offers the potential to diffuse rationalities and ways of thinking well beyond immediate and dyadic relations.

In Fig. 2 the varying tie width between nodes, i.e., the number of shared subsidiaries, stands out. While some firms are tied to each other through joint ownership of a single subsidiary, others hold hundreds of subsidiaries in common, as is shown in the edge weight distribution in Fig. 3. An example is the link between ChemChina and Syngenta, reflecting that the Chinese firm acquired the Swiss agrichemical company Syngenta in 2017. Generally, more ties arguably indicate increased collaboration and/or control, as with the hierarchical relationship between ChemChina and Syngenta, although other important factors are also at play. A tie need not entail an equal ownership share, and such variances might reflect power asymmetries or the differing strategic importance of an entity between its owners. Moreover, if two lead firms share ownership in a major plant, that might not result in a high number of shared subsidiaries, despite being of high importance to the respective

⁵ We use the Chemical and Engineering News' (C&EN) 2020 Global Top 50 ranked according to chemical sales in 2019 and include the two major Chinese chemical producers ChemChina and SinoChem, which do not disclose chemical sales and are not included in the C&EN Top 50.

⁶ These are ChevronPhillips (US), Ecolab (US), and Hanwha (KR) which were ranked respectively 34, 39 and 41 in the C&EN global top 50 for 2020. ChevronPhillips is a joint venture owned by US oil companies Chevron and Phillips 66 and has multiple relevant joint ventures not identified here. For example, the company has recently begun construction of what is set to become the largest ethane cracker in the Middle East together with QatarEnergy (Brelsford, 2024).

Table 2

Sampled companies ranked according to chemical sales.

Company	C&EN description	Headquarters	Ownership	Chemical sales as % of total sales	Chemical sales 2019 (M USD)
BASF	Diversified	Germany	Public	100.0	66,401
Sinopec	Petrochemicals	China	State-owned (China)	14.7	61,596
Dow	Diversified	US	Public	100.0	42,951
SABIC	Petrochemicals	Saudi Arabia	State-owned (Saudi Arabia)	92.4	34,420
INEOS	Diversified	UK	Private	100.0	32,009
Formosa Plastics	Petrochemicals	Taiwan	Public	66.7	31,425
ExxonMobil	Petrochemicals	US	Public	10.7	27,416
Mitsubishi Chemical	Diversified	Japan	Public	83.3	27,353
LyondellBasell	Petrochemicals	US	Public	78.1	27,128
Linde	Industrial gases	UK	Public	90.1	25,429
LG Chem	Diversified	South Korea	Public	100.0	24,554
Air Liquide	Industrial gases	France	Public	98.5	24,171
PetroChina	Petrochemicals	China	State-owned (China)	6.2	22,733
DuPont	Speciality chemicals	US	Public	100.0	21,512
Reliance Industries	Petrochemicals	India	Public	22.0	20,640
Toray Industries	Diversified	Japan	Public	85.4	17,344
Sumitomo Chemical	Diversified	Japan	Public	76.3	15,231
Evonik Industries	Speciality chemicals	Germany	Public	100.0	14,674
Shin Etsu Chemical	Diversified	Japan	Public	100.0	14,158
Covestro	Diversified	Germany	Public	100.0	13,895
Braskem	Petrochemicals	Brazil	Hybrid	100.0	13,267
Lotte Chemical	Diversified	South Korea	Public	100.0	12,973
Yara	Fertilizer	Norway	Hybrid (a)	100.0	12,858
Solvay	Speciality chemicals	Belgium	Public	100.0	12,568
Mitsui Chemicals	Diversified	Japan	Public	100.0	12,282
Hengli Petrochemical	Petrochemicals	China	Private majority	81.2	11,839
Bayer	Agricultural chemicals	Germany	Public	23.6	11,482
Indorama Ventures	Petrochemicals	Thailand	Private	100.0	11,362
Syngenta	Agricultural chemicals	Switzerland	State-owned (China) (b)	78.0	10,588
DSM	Speciality chemicals	Netherlands	Public	100.0	10,086
Asahi Kasei	Speciality chemicals	Japan	Public	50.8	10,027
Wanhua Chemical	Diversified	China	Public	100.0	9851
Arkema	Speciality chemicals	France	Public	100.0	9782
Chevron Phillips Chemical	Petrochemicals	US	Public	100.0	9333
Eastman Chemical	Speciality chemicals	US	Public	100.0	9273
Borealis	Petrochemicals	Austria	Hybrid (Austria & UAE)	100.0	9071
Air Products & Chemicals	Industrial gases	US	Public	100.0	8919
Mosaic	Fertilizers	US	Public	100.0	8906
Ecolab	Diversified	US	Public	59.2	8904
Johnson Matthey	Speciality chemicals	UK	Public	47.4	8819
Hanwha Solutions	Speciality chemicals	South Korea	Public	79.0	8569
Umicore	Speciality chemicals	Belgium	Public	41.9	8196
SK Innovation	Diversified	South Korea	Public	19.1	8186
Westlake Chemical	Petrochemicals	US	Private	100.0	8118
Sasol	Diversified	South Africa	Public	56.7	7995
Nutrien	Fertilizer	Canada	Public	38.6	7729
PTT Global Chemical	Diversified	Thailand	Hybrid (Thailand)	57.6	7660
Lanxess	Speciality chemicals	Germany	Public	100.0	7614
Tosoh	Diversified	Japan	Public	100.0	7211
DIC	Diversified	Japan	Public	100.0	7050
ChemChina	Diversified	China	State-owned (China)	–	–
SinoChem	Diversified	China	State-owned (China)	–	–

Source: Chemical sales data from the Chemical & Engineering News global top 50 for 2020. See [Tullo \(2020\)](#) for details. Notes: Hybrid ownership refers to arrangements with non-majority but significant state ownership. a) The largest shareholder of Yara is the Norwegian government (36.2%) b) Ownership through ChemChina which acquired Syngenta Group in 2020.

Table 3

Descriptive network statistics.

Network	# Nodes	#Edges	Density	Average degree	Weighted average degree	Diameter
One-mode	52	284	0.214	10.9	76.2	4

Note: Density – share of all possible ties that exist in the network, average degree – average number of ties per node, weighted average degree – average number of common subsidiaries per node, diameter – maximum distance between any two nodes in the connected network.

firms. These two aspects are not well captured in the graph (financial data is not available for many of the subsidiaries through ORBIS), but they are relevant to contextualise specific relations within the network.

3.3.2. Network clusters

To capture poly-centricity and identify strongly interconnected clusters in the network, we use an algorithm for community detection, represented by different node colours in [Fig. 2](#) ([Blondel et al., 2008](#)). We identify five clusters in the network, which differ in terms of how strongly they are linked to the rest of the network. A noteworthy cluster is the pink cluster in the upper middle part of [Fig. 2](#), showing strong ownership ties between all Japanese firms in the sample, as well as some other firms mainly from the Republic of Korea. The strong ownership linkages amongst Japanese firms can be attributed to the form and extensiveness of corporate networks in Japan and the institutional logic

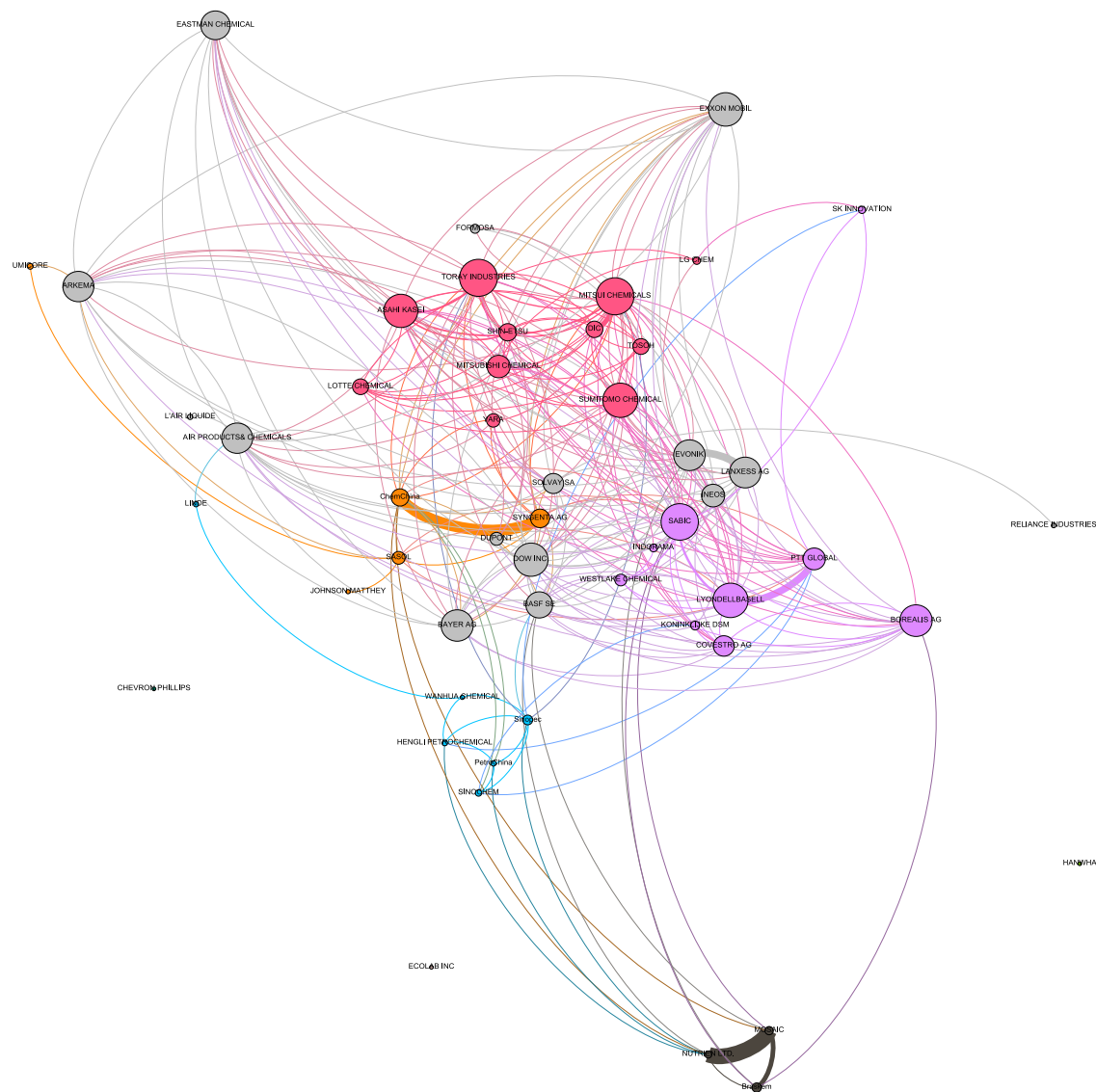


Fig. 2. One-mode ownership network graph. *Note:* Nodes (MNCs) are scaled by eigenvector centrality, while edges (instances of common ownership) are scaled by the number of subsidiaries in which two MNCs share ownership. Colours represent internally strongly connected communities.

governing them (Bhappu, 2000; Morck and Nakamura, 2005). Close collaboration amongst Japanese chemical companies also has historical precedence, most noticeable in the form of a cartel of Japanese chemical firms orchestrated by the Japanese government in the 1970s (Aftalion, 2001). Despite such regionality, the cluster also includes four of the most centrally positioned lead firms in the network, measured by degree count as well as closeness and eigenvector centrality, and is therefore far from isolated from the rest of the network. This integration across scales aligns with the pressure to pursue new markets emphasised by the GPN literature. A recent McKinsey report on the Japanese chemical industry stresses a need for further mergers, acquisitions, and adaptations to global markets to ensure growth (Kakimoto et al., 2023).

While the Chinese producers are hugely important in terms of production capacity (Bauer et al., 2023b), the Chinese cluster in blue appears rather peripheral in the ownership network. The companies in this cluster do not have extensive ties to other actors in the network, despite their substantial and increasing share of global production capacity and status as some of the largest producers worldwide (cf. Bauer et al., 2022). Although several links to other MNCs exist (for example, BASF (DE), PTT Global (THA) and the Japanese cluster), the Chinese firms do not appear amongst the 20 most central actors in the network. In terms

of corporate board and spatial interlocks, Verbeek and Mah (2020) find a similar picture using data from 2018 when assessing relations between 14 lead firms in the petrochemical industry, although Sinopec is the only Chinese firm included in their analysis. Despite this relative isolation in terms of shared subsidiaries, they also find that publicly announced joint ventures between Sinopec and other lead firms are common. All of these joint ventures, though, are located in China (Verbeek and Mah, 2020), reflecting the Chinese industrial policy strategy, which for many years required foreign firms to collaborate with national firms to enter the market⁷ (Howell, 2018). It was through joint ventures with oil and gas majors that petrochemical production capacity skyrocketed in the new millennium because this approach was politically acceptable, enabling expansion while limiting dependence on foreign actors (Tobin, 2019).

While our findings illustrate spatial variation in global regimes in terms of the Chinese focus on self-sufficiency and nationally consolidated corporate networks (Tobin, 2019), this relative isolation does not mean that the ties that do exist cannot strengthen the prospects of trasformismo. Considering betweenness centrality, Sinopec and

⁷ This also reflects that our data and approach do not capture all joint ventures located in China.

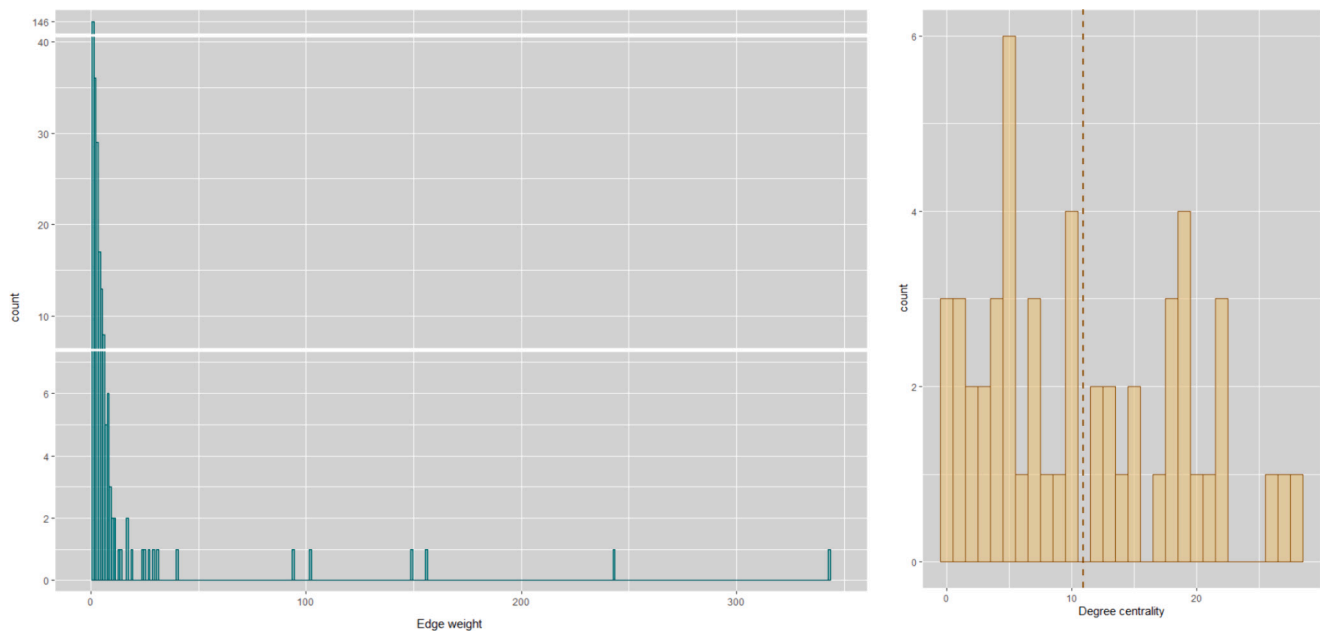


Fig. 3. Network statistics of edge weights and degree centrality in one mode parent x parent network. *Left panel:* Edge weight distribution. Breaks in the vertical axis were added using ggbreak (Xu et al., 2021). *Right panel:* Degree centrality distribution. The dashed line represents the average degree centrality of 10.9.

ChemChina are amongst the ten most central actors, playing a relatively important role in connecting other nodes in the network. Such relations can push actors faced with transition pressures to maintain commitments to fossil feedstock. For example, in the wake of the Russian invasion of Ukraine and increasing gas prices, BASF, the world's biggest chemical company, publicly threatened to “permanently” cut European production while investing in China, building on, and strengthening its existing ties to Chinese producers (Burger, 2022). As such, instead of decreasing fossil-based production in the face of increasing costs, BASF seeks ways in which it can expand its commitments through its existing production network.

4. Reproduction or reconfiguration?

The ownership network mapped above consists mostly of dyadic relations where two lead firms in the network have ownership shares in the same entity (this is the case for 1479 out of 1579 ties). These represent common commitments that we argue underpin investments in fossil-based infrastructure and support trasformismo (see Section 2.2). In this section, we discuss our findings in relation to this argument and consider mechanisms of alignment and accommodation with implications for the prospects for and forms of low-carbon reconfiguration in the petrochemical industry.

4.1. Mechanisms of alignment

The ownership network shows that shared commitments exist between actors working across the fossil fuel value chain. For example, LyondellBasell, the major US-headquartered integrated plastics, chemicals, and refining company, has ties to PTT Global (THA) and SABIC (SAU), both owned by national oil companies, while collaborating with Covestro (DE), a major polymer firm. Similarly, the vertically integrated oil major ExxonMobil (US) notorious for its climate delay and denial efforts (Supran et al., 2023; Supran and Oreskes, 2020, 2021), has ownership interlocks with more downstream, materials-focused firms such as Arkema (FR), Asahi Kasei (JP), and Toray Industries (JP). Ownership interlocks across firms working in related but, in terms of climate mitigation, often thought of as distinct industries, locks in a commitment to a fossil-based global regime across actors in the value

chain from extraction to polymer production, signalling to other industry actors that they commercially and financially devoted, regardless of transition pressures. The network, in that sense, is not only conditioning trasformismo but also an outcome of the existing global regime. The network showcases existing common commitments to fossil-based production, i.e. examples of sites and entities in which actors hold a shared interest in maintaining production at the most profitable capacity until the end of the assets' lifespan. Extensive ownership ties thereby form the basis for promoting fossil-oriented socio-technical reconfigurations and remaining in control of niche, renewable-based technologies, in contrast to alternative pathways focusing on demand change.

In addition to dyadic relationships, the ownership network also illustrates more collective relations and alternative sites of regime diffusion. In the sample, 18 subsidiaries have four or more owners. These include firms managing shared infrastructure, such as gas pipelines, as well as shared think tanks and interest organisations. The most well-connected common-owned entities are policy-oriented and count, e.g., the Singapore Chemical Council (15 ties) and the German BKV (‘think tank of the plastics industry’) (BKV, 2024a) (7 ties). As for the former, Singapore is a long-standing petrochemical hub for refining, transport, and logistics (Breul and Revilla Diez, 2018; Yun and Jin, 2009) and one of the locations where many lead firms are present (Verbeek and Mah, 2020). The Singapore Chemical Council is the ‘official industry association’ and represents the integrated fossil-based energy and chemicals cluster with more than 170 companies (SCIC, 2024). With member companies of different sizes involved in activities along the value chain, from refining to so-called specialty chemicals, the council constitutes a site for local diffusion of the global petrochemical regime. BKV is also supported by lead firms in the network as well as other important industry actors.⁸ Think tanks are important political actors whose ‘coordinative discourses’ influence political processes in and around crises (Hernando et al., 2018; Zimmerman and Stone, 2018). The think tank is thus an example of regime actors joining forces to disseminate policy-relevant knowledge and, by extension, regime alignment.

⁸ Including SABIC, INEOS, BASF, Dow, Covestro, LyondellBasell and Evonik in addition to other important industry actors such as PlasticsEurope (BKV, 2024b).

4.2. Mechanisms of accommodation

In terms of the global regime accommodating transition pressures, the outstanding centrality of SABIC (SAU) in the network is remarkable. With the firm's central location in the global regime and economic direct connections to a host of other actors in the regime, SABIC is in a position to seek alignment around socio-technical reconfiguration and pursue *trasformismo*. SABIC was acquired in 2020 by the national oil company Saudi Aramco, the largest oil producer in the world (SABIC, 2020), and is ultimately owned by the Saudi state. As part of a broader diversification strategy, the consolidation of Saudi Aramco and SABIC encapsulates the belief that petrochemicals will be the future of oil (Tullo, 2019). With new refineries that directly convert crude oil to petrochemicals (Reuters, 2022), the 'preferred technology' for reorienting production from fuels to chemicals (Wu, 2023), the company represents the epitome of an oil-oriented reconfiguration of the industry. In a recent joint venture with Aramco that also includes Chevron as a technology provider, SABIC announced that it was commercializing its crude-to-chemicals technology, seeking to scale up these processes and aiming to convert up to 80% of crude oil into chemicals (Aramco, 2024). In another project, Aramco, together with Dow, is running the largest integrated crude, petrochemical, and plastics complex ever built in a single phase, Sadara, on the basis of massive support from public financial institutions in both Europe and the United States (Skovgaard et al., 2023). The centrality of SABIC therefore signals the continued stability of the global regime in the form of common commitments to expanding fossil-based chemical production.

The centrality of SABIC is also noteworthy in relation to accommodation strategies and the absorption of both landscape and niche pressures. In a coordinated effort with Aramco and Saudi-based actors, SABIC promotes a vision of a future built around 'circular carbon' (Palm et al., 2024). The notion of circular carbon is an effort to reframe commitments to fossil-friendly reconfigurations focusing on carbon capture for utilization and blue hydrogen, allowing for continued oil extraction and petrochemical growth (Shehri et al., 2022). Moreover, the vision of circular carbon is, albeit in different variations, also pushed by European and North American-based actors such as industry organisations and large producers to promote visions of the future that align with growth-oriented strategies (Buck, 2022; Palm et al., 2024). In terms of competing technologies, SABIC has entered partnerships with other lead firms pursuing the development of niche technologies, including electrified steam crackers (BASF et al., 2021). These niche technologies, however, still revolve around the processing of oil and gas and do not fundamentally challenge the dependence on fossil feedstocks. The fact that SABIC, with its state-backed long-term commitment to a fossil-based chemical future, has pursued accommodation of both landscape and niche pressures in tandem with other actors in the network while occupying the most central position, illustrates relations that underpin processes of *trasformismo*.

In opposition to our claim of stability by accommodation and absorption facilitated in and through networks, a well-connected network could arguably also be a vantage point for accelerating sustainability transitions through network diffusion. SABIC, for example, is after all engaging in the development of low-carbon niche technologies with partners in the network. Our point is not that the ownership networks make anything but outright resistance unlikely. Rather, we emphasise here the notion of stability through change. Aligning material interests across otherwise competing actors, ownership relations are part of the structure that can strengthen prospects of *trasformismo* in the face of disruptive change. For example, regardless of origin petrochemical majors use similar arguments and narratives to position themselves as transition enablers (Tilsted et al., 2022). With calls to cap global production and increased attention to the supply side of the plastics crisis (Bergmann et al., 2022), the risks of capital stranding in time of massive expansion—as well as decreased prospects of proliferation through new 'green' markets—constitute real threats to the petrochemical industry

(Mah, 2021b). With varying strength across regions and sites, social movements and forces of 'biojustice environmentalism from below' (Dauvergne and Clapp, 2023), engage in the struggle to shut down petrochemical expansions (Mah, 2023). In this environment, lead firm networks engaging in niche activity work to fend off disruption and strong ownership networks ensure that actors remain materially and juridically committed to fossil-aligned production and solutions.

Regarding capacity additions, Chinese developments are key. A continuation of coal-to-chemicals-based expansions will lead to massive increases in emissions (Cabernard et al., 2021; Li and Hu, 2017; Zhang et al., 2019). In terms of our results, the relative isolation of Chinese producers and the national orientation towards self-sufficiency could indicate less inter-firm pressure to commit to existing assets. Still, all major Chinese producers are state-owned (and in the case of Sinopec and PetroChina, not mainly petrochemical producers), and national priorities are arguably more decisive for petrochemical infrastructure development. At the same time, Chinese large-scale protests against toxic expansions that flourished for years since 2007 have subsided under strengthened societal control (Lee and Ho, 2014; Mah, 2023). Given national dependence on petrochemical production, underpinning domestic economic growth, the prospects for the Chinese cluster to facilitate a sharp break with *trasformismo*-style tactics appear limited. The possibilities for Chinese re-orientation towards low-carbon chemical production require engagement questions of state control, ownership, and national fossil fuel interests (Babić and Dixon, 2022; Nahm and Urpelainen, 2021).

5. Concluding discussion and outlook

In this paper, we have argued that integrated, materially anchored global networks in petrochemicals work to accommodate and adopt pressures, thereby underpinning efforts of *trasformismo*. Starting from the simple observation that ownership signals control and commitment related to direct, dyadic forms of power, the present paper has shown one way of analysing the key role that internationalised networks have for global regimes. Our findings show a truly global yet non-homogenous polycentric network. The lead firm ownership network not only displays economic integration between different corporations; it is also an abstraction that relates to the global petrochemical infrastructure. Many of the subsidiaries in the sample are corporations that are engaged in various manufacturing and distributive activities key to the metabolic organism through which fossil fuels come to play a role in countless aspects of modern life.

Because decisions in the ongoing decade are crucial for the prospects of reaching global temperature goals, the extent to which the global petrochemical regime is currently aligned around pursuing a future for oil in chemicals and fending off disruptive pressures is critical. In our view, the ownership network mapped in this paper constitutes a clear obstacle to low-carbon reorientation. Control and ownership of fossil capital suggest the opportunity for the phase-down and eventually phase-out of assets that are not aligned with ecological limits but also imply a dependency on rents from those very assets. With ownership ties that integrate lead firms along the petrochemical value chain and connect to a fossil fuel-chemicals-plastics nexus, common commitments to existing assets and the centrality of fossil fuel companies, a structure that underpins *trasformismo* towards fossil-based reconfiguration is in place.

Governing the parallel transitions that are needed across sectors involves economy-wide structural transformation with implications that extend far beyond a given focal sector (Andersen and Gulbrandsen, 2020). Because petrochemicals occupy a central position in the fossil energy order and due to the integrated nature of synthetic materials, chemicals, and fossil fuels, there is a stark need for further assessing cross-industry collaboration and the links, integration, and spill-overs along the petrochemical value chain. Many forms of connections between these lead firms exist and the analysis of these parallel connections—which would form a multi-plex network of the same group of

actors—could unpack different forms and mechanisms for isomorphism and alignment with regime rationalities. Relevant network layers could be technology licensing, corporate board interlocks (Buch-Hansen and Larsen, 2021; Verbeek and Mah, 2020), and shared infrastructure (such as pipelines), which directly relate to material system development and decision-making in the network. Looking beyond petrochemical majors, future research could gain from mapping the networks of experts and professionals (Christensen, 2020; Christensen et al., 2020; McCann and Ward, 2010) that (re)produce the global petrochemical regime, including the think tanks and interests organisations also emphasised in this paper. And research related to global financial networks (Coe et al., 2014), and global wealth chains (Seabrooke and Wigan, 2014, 2017), including how they are entangled with global networks of chemical production (Bair et al., 2023), could allow for a better understanding of the financialised nature of fossil-based petrochemicals expansion. Finally, engagement with pressures and counter-pressures and how they vary across regions is crucial to understanding where and how de-escalation to fossil lock-in can arise.

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Appendix A. Descriptive statistics lead firm network (listed by degree count)

Company	Degree	Weighted degree	Betweenness	Closeness	Harmonic closeness	Eigenvector centrality
SABIC	28	128	144.77	0.68	0.78	0.98
Mitsui Chemicals	27	236	93.11	0.69	0.78	1.00
Toray Industries	26	133	71.06	0.68	0.77	1.00
Sumitomo Chemical	22	69	27.11	0.62	0.72	0.92
Asahi Kasei	22	128	39.85	0.61	0.71	0.89
LyondellBasell	22	243	33.51	0.62	0.72	0.92
Exxon Mobil	21	28	27.57	0.62	0.71	0.89
Dow Inc.	20	67	16.50	0.61	0.70	0.89
Borealis Ag	19	29	17.12	0.58	0.68	0.84
Evonik	19	196	41.69	0.59	0.68	0.81
BASF SE	19	66	59.36	0.59	0.68	0.67
Air Products & Chemicals	19	28	98.53	0.59	0.68	0.80
Bayer Ag	18	34	8.80	0.59	0.67	0.83
Lanxess Ag	18	189	15.69	0.57	0.67	0.81
Arkema	18	21	43.17	0.57	0.67	0.80
PTT Global	17	184	85.17	0.58	0.66	0.54
Mitsubishi Chemical	15	65	18.78	0.55	0.63	0.56
Eastman Chemical	15	15	1.07	0.55	0.63	0.75
INEOS	14	50	16.28	0.55	0.63	0.57
Solvay Sa	13	37	53.12	0.53	0.61	0.49
ChemChina	13	268	52.23	0.54	0.61	0.41
Syngenta Ag	12	276	17.12	0.55	0.61	0.45
Covestro Ag	12	29	11.55	0.52	0.60	0.50
Shin-Etsu	10	100	1.40	0.51	0.58	0.41
Tosoh	10	57	4.83	0.49	0.56	0.36
Dic	10	129	1.86	0.49	0.57	0.38
Lotte Chemical	10	35	5.33	0.48	0.56	0.36
Sinopec	9	22	57.99	0.51	0.57	0.18
Sasol	8	16	59.53	0.49	0.55	0.27
Nutrien Ltd.	7	351	12.55	0.42	0.49	0.10
Yara	7	24	1.81	0.49	0.54	0.29
Dupont	7	33	3.69	0.47	0.53	0.28
Westlake Chemical	6	38	2.55	0.46	0.51	0.24
Mosaic	5	440	2.18	0.47	0.51	0.14
Braskem	5	99	4.52	0.45	0.50	0.17
Hengli Petrochemical	5	13	13.21	0.41	0.47	0.05
Petrochina	5	12	2.68	0.39	0.45	0.05
Sinochem	5	7	9.31	0.44	0.49	0.08
Koninklijke Dsm	5	19	7.31	0.47	0.51	0.15
Formosa	4	5	1.37	0.44	0.48	0.16

(continued on next page)

CRediT authorship contribution statement

Joachim Peter Tilsted: Conceptualization, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Fredric Bauer:** Funding acquisition, Investigation, Methodology, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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(continued)

Company	Degree	Weighted degree	Betweenness	Closeness	Harmonic closeness	Eigenvector centrality
LG Chem	4	8	1.86	0.42	0.47	0.11
SK Innovation	4	4	7.28	0.47	0.51	0.11
Wanhua Chemical	3	11	5.95	0.36	0.40	0.02
Indorama	3	9	0.44	0.44	0.48	0.12
Umicore	2	2	0.00	0.38	0.42	0.06
Linde	2	2	7.19	0.40	0.43	0.05
L'air Liquide	1	2	0.00	0.37	0.40	0.05
Johnson Matthey	1	2	0.00	0.33	0.36	0.02
Reliance Industries	1	1	0.00	0.35	0.38	0.03
Hanwha	0	0	0.00	0.00	0.00	0.00
Chevron Phillips	0	0	0.00	0.00	0.00	0.00
Ecolab Inc	0	0	0.00	0.00	0.00	0.00

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